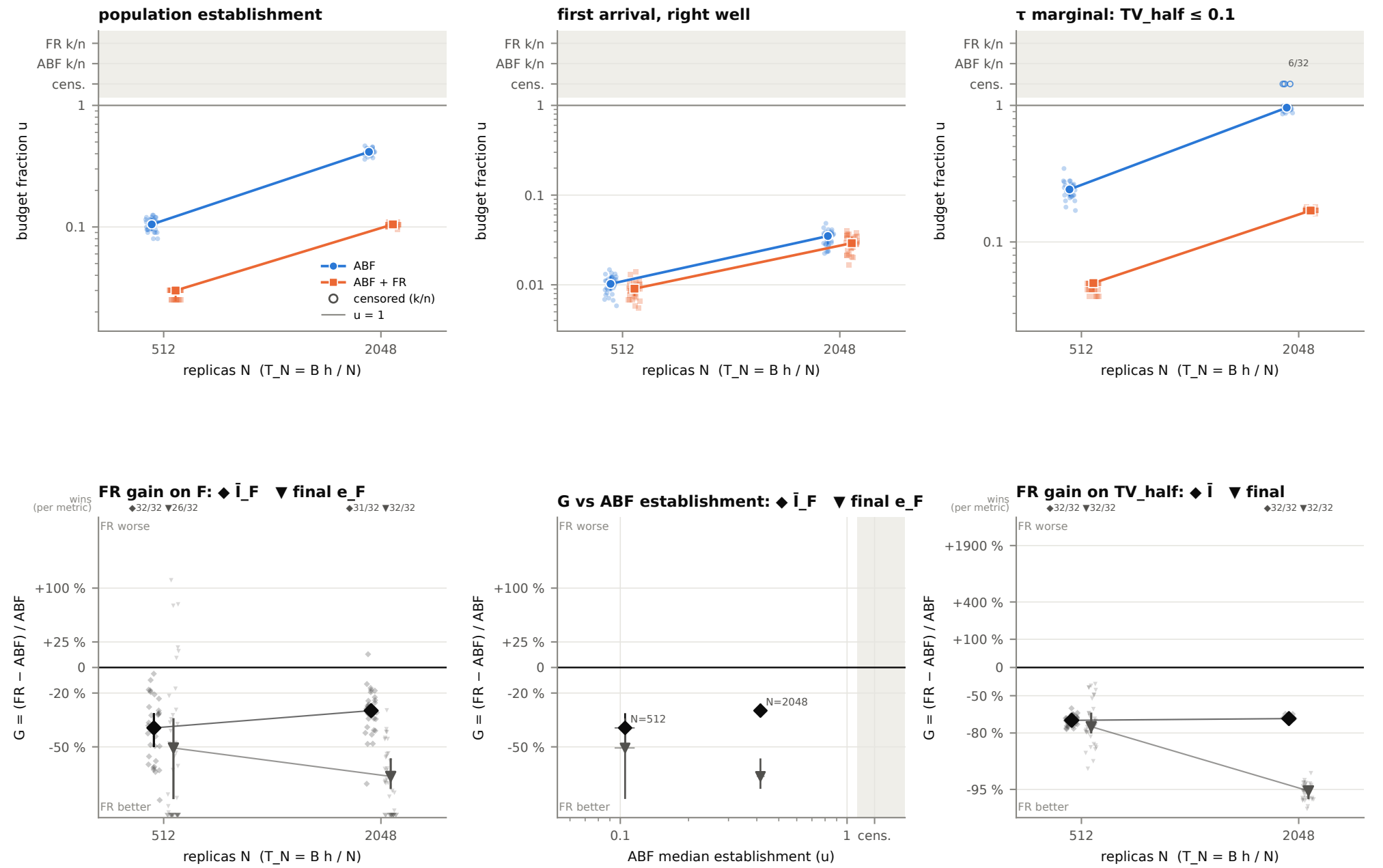


gateway lam1: population establishment, first arrival and marginal convergence vs N (budget fraction)

Equal total budget per arm per seed:  $B = N \times n\_steps = 3.2768e+09$  walker-steps at every  $N$  ( $h = 2.5e-05$ ,  $T\_N = B \cdot h / N$ ); 32 seeds per ( $N$ , method). Do FR gains coincide with establishment-limited  $N$ ?



gateway lam1 · S6a\_establishment\_budget · status: all planned ( $N$ , seed, method) complete · FR is not defined at  $N = 1$  (ABF only, by design).  
Establishment = first save after which the instantaneous far-state fraction (right well  $x > 0.5$ ) stays  $\geq \frac{1}{2} \times 0.361$ . Shaded  $N$  [ $]$ : the instantaneous rule is UNRELIABLE there (ill-defined at  $N \leq 2$ ; an exactly uniform population passes every save of the second half with probability  $< 0.9$ , summary establishment\_null), so the same rule on the cumulative visitation is plotted; FR birth-death can push the instantaneous count below binomial scatter, so an FR establishment advantage at those  $N$  is not evidence of faster convergence.  $\tau(TV\_half)$  on the budget axis uses the common budget grid.  
Top: seed median (large open marker in the 'cens.' band = median censored), IQR bars, faint seeds. Bottom: paired  $G$  with bootstrap 95 % CIs (filled = excludes 0; 'wins' =  $FR < ABF / n$ ); scatters: each  $N$ 's  $G$  against the ABF median time (horizontal bar = ABF IQR; 'cens.' = censored).  
Source results/mechanism/conditional\_relaxation/lam1/analysis/summary.json (generated 2026-10-10T14:19:15Z, eqb\_metrics/2, 10000 bootstrap resamples); plot\_synthesis/2.